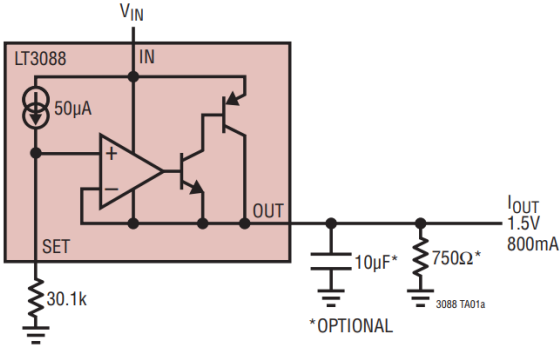


LT3088



Please walk through the datasheet **LT3088** to answer the following questions:

Google: **LT3088 datasheet**

Q1. What is the input **voltage range** of this regulator?

Q2. What is the output voltage range of this regulator?

Q3. How much **maximum output current** this device can provide?

Q4. What is the set pin current of this regulator?

Q5. What packages are available for this regulator, and how to identify the package from the part number?

Q6. To which node of this device should we connect R_{SET} ?

Q7. What is a minimum load current required for the regulation of the output voltage?

Q8. Why use C_{SET} and C_{OUT} , and what are the appropriate value for these capacitors?

Q9. If $R_{SET} = 30.1 \text{ k}\Omega$, calculate the output voltage. If $C_{SET} = 100\text{nF}$ is added in parallel with R_{SET} , determine the time required for the output to stabilize, and verify the result using an LTspice simulation?

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